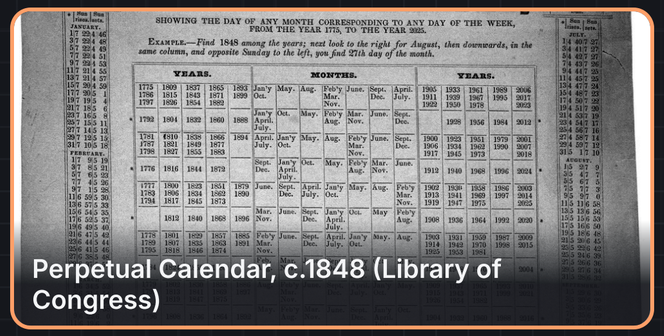
Pi & Planck

CALENDAR MATH

Friday the 13th: The Numbers

Is the 13th really more likely to fall on a Friday? Yes, by a hair.

A CALENDAR THAT REPEATS


SHOWING THE DAY OF ANY MONTH CORRESPONDING TO ANY DAY OF THE WEEK, FROM THE YEAR 1775, TO THE YEAR 2825.

EXAMPLE.—*Find 1848 among the years; next look to the right for August, then downwards, in the same column, and opposite Sunday to the left, you find 27th day of the month.*

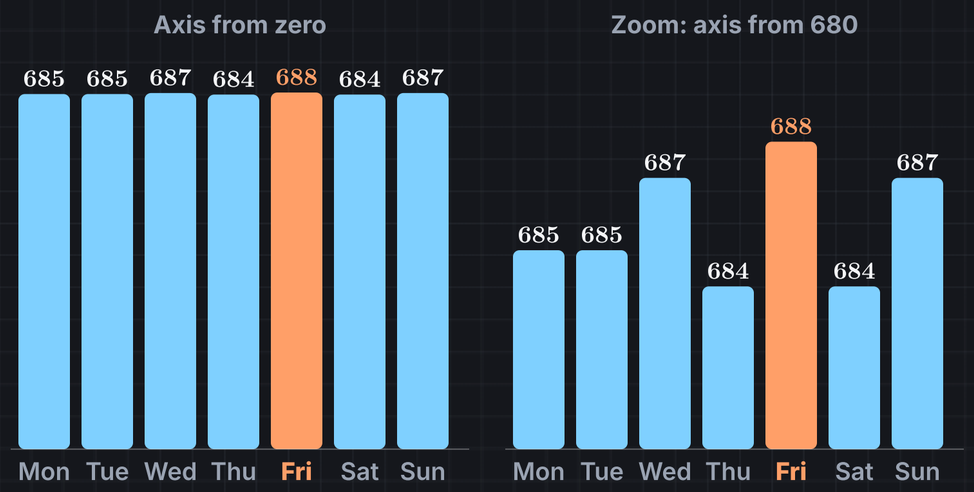
Perpetual Calendar, c.1848 (Library of Congress)

1582 Pope Gregory XIII's bull Inter gravissimas sets the leap-year rule we still use.

97 leap days in every 400 years.

146,097 days in 400 years: exactly 20,871 weeks.

THE 13TH, WEEKDAY BY WEEKDAY (400 YEARS)



4,800 thirteenths in 400 years. An even split would be 685.7 each. Friday gets 688.

THE WHY

1. The calendar repeats exactly every 400 years.
2. So the 4,800 thirteenths cannot split evenly into 7 weekdays.
3. Friday gets the extra: 688, against 684 for Thursday and Saturday.

ONE YEAR HAS 1 TO 3

Jan	Feb	Mar	Apr	May	Jun
Jul	Aug	Sep	Oct	Nov	Dec

1 Friday the 13th in 171 of 400 years, 2 in 170, 3 in 59. In 2026: three.

688 in 400 years — Fridays that are the 13th. The most of any weekday, first observed by B. H. Brown in 1933.

14.33% vs 14.29% — Friday's share of thirteenths, against an even one in seven.

1 to 3 per year — Friday the 13ths in any year. 2026 has three.

Sources: E. W. Weisstein, Friday the Thirteenth, Wolfram ScienceWorld · B. H. Brown, Amer. Math. Monthly 40 (1933) 607 · L. Doggett, Calendars and their History, NASA · US Naval Observatory, Leap Years · A. Ziggelaar, The Papal Bull of 1582 · Counts verified by code · Image: Library of Congress

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